

Shadow Regulatory Citation Map v1.1

Regulatory Citation × Persona × Test Map

Version: 1.1 · **Generated:** 2026-07-06 · **Last updated:** 2026-07-07 (4-layer procurement structure + CSV export + CI Python 3.9-3.13) · **Framework:** SR 26-2 Tier 3 companion control (effective 2026-04-17)

Authority: - Loredana C. Levitchi — BRD + Addenda A/B/C banking domain grounding, regulatory citation review, 4-layer procurement structure (2026-07-06 Test Stack contribution) - Alex Xiaoyu Ji — schema mapping, test coverage extraction, cryptographic attestation binding

Purpose: This document is the procurement-audience map from Shadow's runtime personas to (1) the specific regulatory citations each voice is tested against and (2) the exact test files that exercise each citation. Bank counsel opening the repo can trace a regulatory obligation to a Shadow persona to a runtime test, all in one traversal.

How to read this document: every row is a triple `<persona, citation, test file>` and represents a defended invariant. If a bank auditor asks *"does Shadow's Fair Lending Compliance voice test the CFPB Bulletin 2024-09 model-traceability requirement?"* the answer is row-lookup, not code archaeology.

Also available as `docs/CITATION_MAP.csv` for procurement counsel who want to filter / sort / import into a GRC system.

0. Recommended procurement format — 4-layer structure

Structure contributed by Loredana C. Levitchi (2026-07-06 Test Stack package).

Bank counsel who want to audit Shadow's regulatory posture should read the material in four layers, top-to-bottom. Each layer answers a specific counsel question and points to the underlying artifact.

Layer 1 — Persona-to-Regulation Mapping. *"Which Shadow voice tests which regulation?"* Read Section 2 of this document (or `docs/CITATION_MAP.csv` if you prefer a spreadsheet). Every row is `<persona, citation, control objective, exact test file, assertion>`. Nothing is aspirational — every test named here runs in CI on every merge.

Layer 2 — Canonical Attestation Regression. *"Is the signed record I'm reviewing untampered?"* Every Shadow decision produces an Ed25519-signed attestation (RFC 8032) with a canonical payload defined in `lib/attestation.js`. The same canonical payload is re-verified byte-for-byte by the Python `shadow_verify` library on Python 3.9-3.13 in CI. Cross-language drift is the load-bearing invariant — see `test/python-verify-cross-lang.test.js` + Python `shadow-verify/tests/`.

Layer 3 — Evidence Binder. *"Where is the reproducible proof that CI passed for this specific commit?"* GitHub Actions test log for the pinned commit + npm test output + pytest output + `benchmark/history/SUMMARY.md` for the Shadow Agentic Score history. Every regression in a Layer 1 test blocks the merge gate, so the evidence binder is not aspirational — it is enforced by branch protection.

Layer 4 — Compliance Narrative. *“Why does each persona exist and what regulatory risk does it control?”* Read Section 1 (Persona Overview) below. Each persona has one role, one AA-code assignment, and one L3 source document. If counsel wants a longer narrative, see `docs/BRD_ALIGNMENT.md` for the source separation principle.

The 4-layer structure gives counsel four different navigation paths for four different audit questions. Reading only Layer 1 is enough for a rubric-level compliance review. Reading Layer 1 + Layer 4 is enough for procurement approval. Layers 2 + 3 exist for the deeper technical audit that follows sign-on.

1. Persona Overview

Persona	Role	AA Codes	L3 Source Document
Credit Fundamentals	Underwriter — evaluate borrower creditworthiness against institutional floors	AA01, AA02	Addendum A (Credit Policy) + Addendum B (DTI Policy)
Risk Officer	Portfolio VaR + concentration + LTV — institutional-side risk appetite	AA03, AA04	Addendum C (LTV Policy) + Risk Appetite Note
Fair Lending Compliance	ECOA / Reg B disparate-impact + adverse-action notice compliance	AA05	BRD Governance Controls (ECOA / Reg B integration)
Customer Advocate	Adverse-action explanation quality + borrower-facing readability	(escalates all)	CFPB Bulletin 2024-09 (model traceability)
Macro Contrarian	Sector cycle + recession sensitivity — devil’s advocate	(none)	Institutional risk appetite note (sector-cycle overlay)
AML/KYC Investigator (opt-in)	AML / Sanctions / KYC — regulatory eligibility	AA06	BSA + USA PATRIOT §326 + FinCEN CDD (31 CFR 1010.230) + OFAC 50% rule

The AML/KYC voice is opt-in: it attaches only when the loan payload contains `aml_flags[]` or `kyc_status` fields (per `lib/aml-kyc-voice.js` `AML_FLAG_POLICY` + `KYC_STATUS_POLICY`).

2. Citation × Persona × Test File Map

2.1 US federal banking supervision

Citation	Full name	Persona(s) enforcing	Primary test file	Test count
SR 26-2	Federal Reserve Model Risk Management Guidance (formerly SR 11-7, replaced 2026-04-17)	All 6	test/traceability-and-guardrail.test.js	19
SR 11-7 (rescinded)	Predecessor to SR 26-2 — retained in lib/traceability.js:33 as historical annotation for legacy audit trails	All 6	test/traceability-and-guardrail.test.js	19
CFPB Circular 2022-03	Requires denials to cite the specific policy source, no template phrases	Fair Lending + Customer Advocate	test/reason-code-dictionary.test.js	17

2.2 Fair lending / ECOA / Reg B (Regulation B)

Citation	Full name	Persona	Primary test file(s)	Test count
ECOA	Equal Credit Opportunity Act — prohibited-basis analysis	Fair Lending Compliance	test/run-loan-council.test.js	15
Reg B (12 CFR 1002)	ECOA implementing regulation — adverse action notice requirements	Fair Lending Compliance	test/reason-code-dictionary.test.js + test/run-loan-council.test.js	17 + 15
Reg B adverse-action codes AA01-AA06	Signed reason-code dictionary — post-hoc dictionary swap breaks attestation verification (v1.5.8)	Fair Lending Compliance + Credit + Risk + AML/KYC	test/dictionary-hash-binding.test.js + test/reason-code-dictionary.test.js	(bound) + 17
CFPB Bulletin 2024-09	Model-traceability for adverse-action explanations	Customer Advocate	test/traceability-reproducibility.test.js	14
Protected-class proxy blocklist	15-item ECOA proxy attribute blocklist enforced in lib/enforce-reason-code-dictionary.js	All 6	test/reason-code-dictionary.test.js	17

2.3 AML / KYC / Sanctions (opt-in AML/KYC voice)

Citation	Full name	Persona	Primary test file(s)	Test count
BSA (Bank Secrecy Act)	31 USC 5311 — recordkeeping + reporting	AML/KYC Investigator	test/aml-kyc-voice.test.js	24
BSA §5318(g)(2) — tipping-off	Borrower-facing rationale must NOT name specific SAR-related AML flag	AML/KYC Investigator	test/aml-kyc-adversarial.test.js	23
USA PATRIOT Act §326	Customer Identification Program (CIP) — CDD verification	AML/KYC Investigator	test/aml-kyc-voice.test.js	24
FinCEN Customer Due Diligence rule	31 CFR 1010.230 — beneficial ownership + risk-based CDD program	AML/KYC Investigator	test/aml-kyc-voice.test.js	24
OFAC SDN list match	Sanctions program — Specially Designated Nationals list	AML/KYC Investigator	test/aml-kyc-adversarial.test.js	23
OFAC 50% rule	Sanctions ownership aggregation	AML/KYC Investigator	test/aml-kyc-voice.test.js	24
FinCEN 2026-Q1 GTOs	Geographic Targeting Orders — extended coverage	AML/KYC Investigator	test/aml-kyc-voice.test.js	24
FATF Recommendation 10	International CDD framework — risk-based approach	AML/KYC Investigator	test/aml-kyc-voice.test.js	24

2.4 EU credit-scoring regime

Citation	Full name	Persona	Primary test file(s)	Test count
GDPR Article 22	Automated individual decision-making + right to meaningful information	Fair Lending + Customer Advocate	test/reason-code-dictionary.test.js	17
ECJ Schufa case (C-634/21)	2023 ruling — credit scoring qualifies as "automated decision" under Article 22	Fair Lending + Customer Advocate	test/traceability-reproducibility.test.js	14
EU AI Act Article 14 — human oversight	High-risk system oversight — audit trail export requirements	All 6	test/attestation-acceptance-demo.test.js	(chain)
EU AI Act — credit-scoring deferral to 2027-12-02	Digital Omnibus Regulation 2026-05 pushed credit-scoring deadlines	(context only)	(documented in docs/EU_AI_ACT_STATUS.md)	N/A

2.5 Investor conduct / suitability (advisor pack)

Citation	Full name	Persona	Primary test file(s)	Test count
Reg BI (Regulation Best Interest)	Broker-dealer suitability standard — fiduciary verdict engine	(advisor pack, roadmap)	test/persona-schema.test.js (schema anchor test)	14
FINRA Rule 2111	Suitability requirements	(advisor pack, roadmap)	(persona schema anchor: Reg BI)	(via schema)

2.6 Cryptographic attestation invariants

Citation	Full name	Persona-independent	Primary test file(s)	Test count
RFC 8032 (Ed25519)	Edwards-Curve Digital Signature Algorithm — SR 26-2 Tier 3 attestation binding	(all decisions)	<code>test/attestation-ed25519.test.js</code> + <code>test/attestation.test.js</code>	13 + 20
Dictionary-hash binding (v1.5.8)	Reason-code dictionary SHA-256 signed into every attestation payload — post-hoc dictionary edit breaks verification	(all decisions)	<code>test/dictionary-hash-binding.test.js</code>	(dedicated file)
Hash-chain integrity (v1.5.10)	<code>previous_hash</code> chain — detects reordering / insertion / truncation / edit-cascade	(all decisions)	<code>test/attestation-chain.test.js</code>	22
Schema-runtime coherence (v1.5.11)	Persona L1/L2/L3 thresholds pinned against runtime <code>LOAN_DEFAULTS</code> — silent prompt drift blocked	All 6	<code>test/schema-runtime-coherence.test.js</code>	48
Cross-language Node ↔ Python verifier	<code>shadow_verify</code> Python library replicates Node signing payload byte-for-byte	(all decisions)	<code>test/python-verify-cross-lang.test.js</code>	(cross-lang)

2.7 Adversarial / red-team invariants

Citation	Full name	Persona	Primary test file(s)	Test count
MCPTox canary tokens (arXiv:2508.14925)	Prompt-injection detection at council output boundary	All 6	test/mcptox-canary.test.js	(dedicated)
Free-MAD hallucination amplification (arXiv:2509.11035)	Provider diversity — never silently substitute providers during diverse routing	All 6	test/provider-diversity.test.js	18
Hidden-anchor bias (arXiv:2606.19494)	Randomize persona presentation order to defeat position bias	All 6	test/presentation-order.test.js	12
Analysis-only guardrail	12-pattern regex catches hallucinated trade-execution verbs at council output	All 6	test/traceability-and-guardrail.test.js	19

3. Persona × Citation Coverage Matrix

Persona	US federal (SR 26-2, CFPB)	EOCA / Reg B	AML / KYC / OFAC	EU (GDPR, Schufa)	Reg BI	Ed25519 / Hash- chain	Adversarial
Credit Fundamentals	✓	✓	—	✓ (via reason codes)	(roadmap)	✓	✓
Risk Officer	✓	✓ (LTV via AA03/AA04)	—	✓	(roadmap)	✓	✓
Fair Lending Compliance	✓	✓ (primary)	—	✓ (primary)	(roadmap)	✓	✓
Customer Advocate	✓ (Bulletin 2024-09)	✓	—	✓ (Schufa case)	(roadmap)	✓	✓
Macro Contrarian	✓	—	—	—	(roadmap)	✓	✓
AML/KYC Investigator (opt-in)	✓	✓ (AA06)	✓ (primary)	—	—	✓	✓

Legend: ✓ = covered by named runtime test; — = not applicable to persona role; (roadmap) = advisor pack shipping post-Reg BI vertical launch.

4. Test count summary

- **Total runtime tests:** 668 (v1.5.11)
- **AML/KYC coverage (voice + adversarial):** 47 tests (24 + 23)
- **Attestation cryptographic coverage:** 55+ tests (Ed25519 + hash-chain + dictionary-hash binding + cross-language)
- **Reg B reason-code dictionary coverage:** 17 tests (`test/reason-code-dictionary.test.js`)
- **Persona schema × runtime coherence gate:** 48 tests (`test/schema-runtime-coherence.test.js`)
- **Provider diversity + hallucination amplification defense:** 18 tests (`test/provider-diversity.test.js`)

Full test surface: run `npm test` at repository root. All 668 tests currently pass, zero regressions across v1.5.7 to v1.5.11.

5. Reviewer notes

For bank counsel / regulatory examiner: the *citations* in Section 2 are the regulatory obligations. The tests named in each row are the deterministic invariants Shadow ships to defend that obligation at runtime. A regression in any named test blocks the CI merge gate, so the coverage is not aspirational — it is enforced.

For grad-school co-authors / academic collaborators: this document is a live-updated appendix to any paper citing Shadow's runtime coverage. Please add a citation to `docs/CITATION_MAP.md` (this file) in the paper's methods section so reviewers can trace the empirical claim to the specific test file.

For Lora specifically: every row in Section 2 flows from your BRD + Addenda A/B/C. The rows we discussed on 2026-06-19 (FICO 700 hard block, DTI 0.36 escalate, Tier 3 SR 26-2 companion positioning) are all in Section 2.1 and 2.2. The dictionary-hash binding (v1.5.8) closes the Reg B post-hoc dictionary edit gap you flagged. The AML/KYC voice (v1.4.0) is opt-in per your July 2026 feedback that BSA §5318(g)(2) tipping-off must NOT be triggered when the loan payload doesn't declare `aml_flags[]` or `kyc_status`.

If any row is *missing*, mis-classified, or needs a stronger citation — please annotate this file directly and I will iterate. This map is the artifact bank auditors will read before they read the code.

6. Contact & feedback

- Alex Xiaoyu Ji · xji1@mail.yu.edu · schema mapping + test coverage + cryptographic binding
- Loredana C. Levitchi · regulatory citation authority + BRD + Addenda A/B/C

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